Microcontroller-Based Voltage Protector Device

1. Introduction

The Microcontroller-Based Voltage Protector Device is designed to protect electrical appliances from abnormal supply voltage. Overvoltage and undervoltage can damage electronic equipment and reduce its lifespan. My project uses an Arduino Uno to monitor voltage and control the power supply to a connected load.

2. Design Approach

The system was designed using an Arduino Uno, a step-down transformer, a voltage-sensing circuit, a 16×2 LCD, a 5V relay, resistors, a capacitor, and an LED. The transformer reduces the AC voltage for sensing, while the sensing circuit converts it into a suitable signal for the microcontroller. The Arduino compares the measured voltage with predefined limits and controls the relay accordingly.

3. Implementation Process

First, I connected the transformer and voltage-sensing circuit to the Arduino's analog input. Next, I interfaced the LCD to display voltage readings and system status. The relay was connected to control the electrical load, and an LED was included for visual indication. Finally, the Arduino was programmed to monitor the voltage continuously and disconnect the load whenever the voltage exceeded the specified range. The transformer steps down the AC supply voltage, and the voltage-sensing circuit converts it into a suitable signal for the Arduino Uno. The microcontroller continuously monitors the input voltage and compares it with the predefined safety limits. For this demonstration, the normal operating range is set to 180–250 V AC. If the voltage falls below 180 V AC or rises above 250 V AC, the Arduino activates the relay to disconnect the electrical load. The 16×2 LCD displays the voltage and protection status, while the LED provides a visual indication of the system's operating condition. When the voltage returns to the safe range, the load can be reconnected according to the programmed recovery logic.

